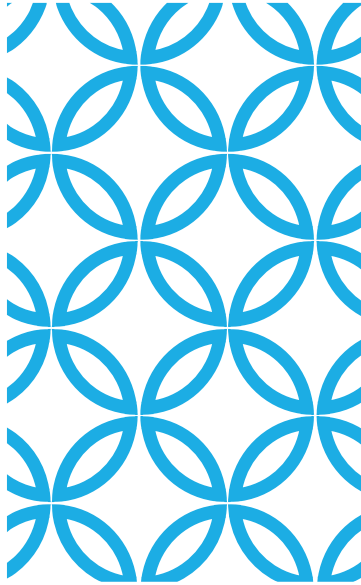




HORMONE SYNTHESIS

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Learning Objectives

1. Compare in general terms the typical plasma levels of hormones with those of nutrients and proteins.
2. List the classes of hormones that are subject to protein binding in the blood.
3. State the importance of plasma protein binding for hormone action.
4. List the precursors for amino acid derived hormones, steroid hormones and eicosanoid hormones.
5. Relate the synthesis and half lives of insulin to C-peptide.
6. Compare the synthesis of oxytocin and vasopressin (ADH).
7. Name the biosynthetic precursor of ACTH, b-lipotropin, MSH and β -endorphin.
8. Describe the proteolytic steps in angiotensin II activation.

Outline:

1. Classes of hormones
2. Blood levels of hormones
3. Plasma transport via carrier proteins
4. Amino acid-derived hormones
5. Steroid hormones
6. Eicosanoids
7. Peptide hormones



Classes of Hormones

1. Amino acid derived hormones

- Catecholamines (e.g., epinephrine, norepinephrine)
- Serotonin, melatonin
- Thyroxine (T_4) and triiodothyronine (T_3)

2. Steroid and lipid derived

- Mineralocorticoids (aldosterone)
- Glucocorticoids (cortisol)
- Androgens and estrogens
- Eicosanoids: prostaglandins, thromboxane, leukotrienes

3. Protein/Peptide

- Insulin, glucagon, growth hormone
- Vasopressin (ADH), oxytocin

Blood Levels of Hormones

Some hormones are bound to binding proteins/carriers in blood:

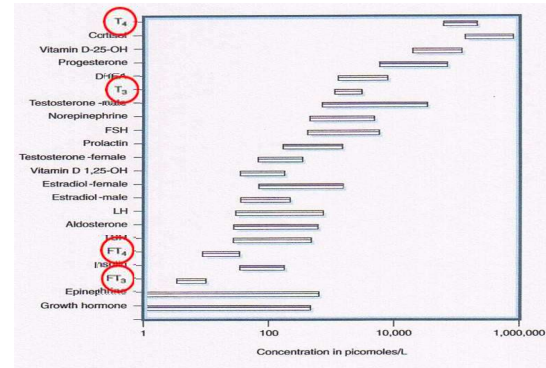
- Reduced activity
- Increased half-life

Free hormone as active form:

- Binds receptors as free
- Cross into interstitium from blood vessels and/or into cells
- Reduced half-life

Blood Levels of Hormones

- When measuring blood concentration
 - **Total concentrations**
 - less sensitive to hormone action or secretion
 - **Bound concentrations**
 - Relatively constant
 - Generally proportional to total hormone concentrations
 - *Easiest to measure for most hormones*
 - **Free concentrations**
 - Typically, the physiologically relevant form
 - Sensitive to changes in use or secretion
 - Hardest to measure for many hormones



Circulating level of hormones vary. Note the low circulating levels of T3 and T4.

Carrier Proteins for Plasma Transport

Many hormones must be carried through blood

- Some binding proteins are specific
 - **Thyroxine binding globulin (TBG)** transports thyroxine in blood
 - **Growth hormone (GH)** is transported by **GH binding protein (GHBP)**
- Transport globulins bind a hormone class for blood transport
 - **Steroid hormones** are transported on albumin or specific globulin transport proteins
 - **Transcortin binds** progestins and glucocorticoids in blood

Carriers provide:

- Protection from destruction in blood
- Solubility in the blood
- Limit release into tissues (increases half life in blood)
- Increase hormone pool and evens out spikes in secretion

The growth hormone-binding protein is a soluble, extracellular and circulating protein which, in humans, is generated by proteolytic cleavage of the cell membrane-bound growth hormone receptor.

Given its structural relationship to the extracellular hormone-binding domain of the growth hormone receptor, it retains the ability to bind growth hormone with high affinity and has the potential to modulate the diverse actions of growth hormone – either by increasing the half-life of GH in the circulation, by sequestering GH away from its cellular receptor or, through potential dimerization with the full-length receptor, acting as a dominant-negative regulator of receptor signaling.

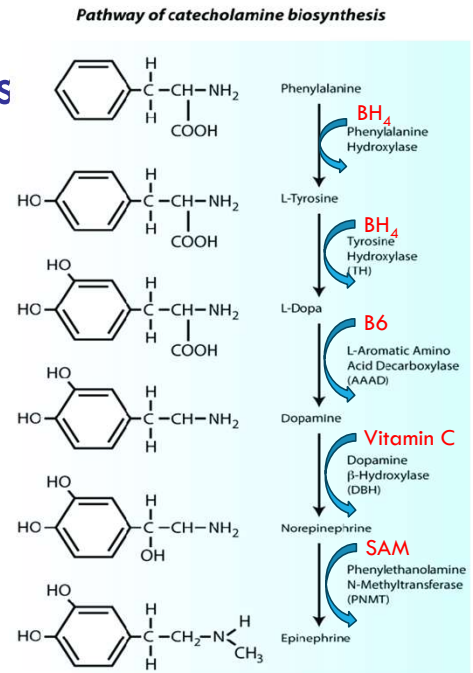
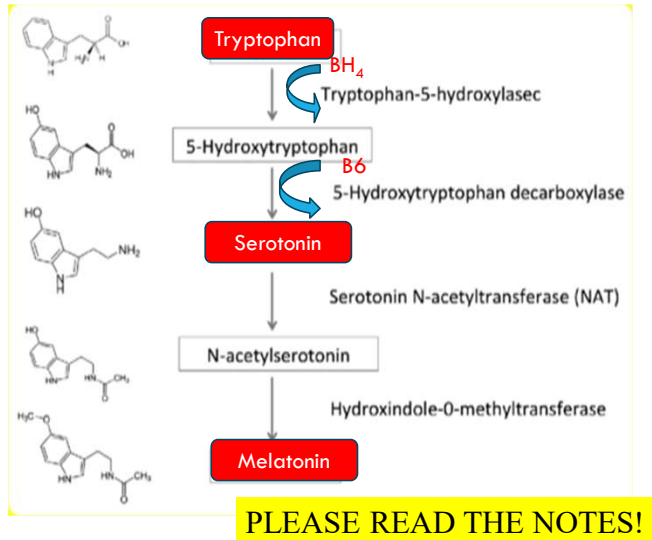
Sex hormone-binding globulin (SHBG) transports androgens and estrogens in blood and regulates their access to target tissues. Hepatic production of SHBG fluctuates throughout the life cycle and is influenced primarily by metabolic and hormonal factors.

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Amino Acid-Derived Hormones



1. Synthesis and Significance of 5-HT:

• Also called “Serotonin”.

• There are three main types of cells that store serotonin (5-hydroxytryptamine) -

- **Blood platelets**
- **Neurons in the brain and the intestinal myenteric plexus,**
- **Mucosa of the gastrointestinal tract contains enterochromaffin cells.**

• The enterochromaffin cells and serotonergic neurons produce serotonin from L-tryptophan,

• Platelets uptake the serotonin from blood.

• In the cytosol of brain cells, the enzyme L-tryptophan hydroxylase (TPH) is involved in the synthesis of serotonin through the conversion of L-tryptophan to 5-hydroxytryptophan.

• Serotonin synthesis in neurons is regulated by this enzyme in a similar fashion to that by the related enzyme L-tyrosine hydroxylase, which converts L-tyrosine to L-dihydroxyphenylalanine (L-DOPA).

•**Functions of Serotonin:**

- **Mood elevator.**
- **Involved in bone metabolism.**
- **Involved in cardiovascular health.**
- **Involved in eye health.**
- **Involved in blood clotting.**
- **Involved in neurological disorders.**

2.Melatonin:

•**Secreted by “Pineal Gland” and regulates sleep.**

•**Secretion depends on light, secreted in dim light i.e. during night, effects are minimum during day.**

•**Synthesized in the body from a similar pathway of Serotonin.**

•**Functions:**

- Sleep quality and mood are improved.
- Melatonin functions can delay aging processes.
- Scavenging free radicals,
- Stabilizing biological rhythms,
- Stimulates the immune system.
- Decreases susceptibility to stress.

As a dietary supplement, melatonin is often used for the short-term treatment of insomnia, such as from jet lag or shift work, and is typically taken by mouth.

Serotonin N-acetyl Transferase (SNAT) controls the daily rhythmic production of melatonin.

3.Catecholamines:

•The substances containing **catechol nuclei** in their chemical structure are called “**Catecholamines**”

•Following are important catecholamines in the body,

- **Dopamine.**
- **Adrenaline.**
- **Noradrenaline.**

•**Tyrosine is the precursor for the synthesis of catecholamines.**

•The conversion of **tyrosine to catecholamines** occurs in **adrenal medulla and central nervous system.**

•*Synthesis of Catecholamines:*

•Tyrosine is hydroxylated to 3,4-dihydroxyphenylalanine (DOPA) by tyrosine hydroxylase.

•This reaction requires tetra hydro biopterin as coenzyme.

•DOPA undergoes PLPdependent decarboxylation to give dopamine which, in turn, is hydroxylated to produce norepinephrine.

•Methylation of norepinephrine by S-adenosyl methionine gives epinephrine.

•The difference between epinephrine and norepinephrine is only a methyl group (norepinephrine has no methyl group).

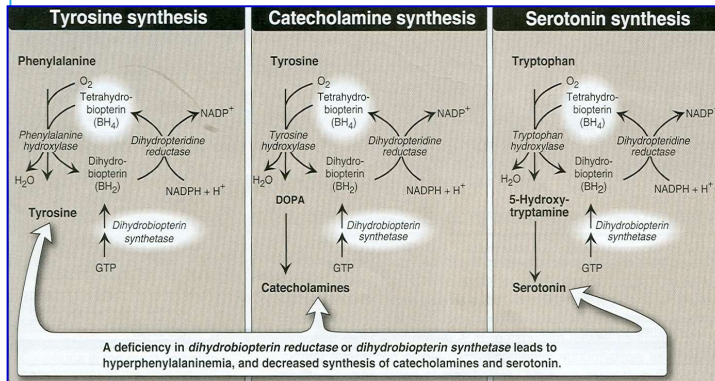
•*Functions of catecholamines :*

- Norepinephrine and epinephrine regulate carbohydrate and lipid metabolisms.
- They stimulate the degradation of triacylglycerol and glycogen.
- They cause an increase in blood pressure.
- Dopamine and norepinephrine serve as neurotransmitters in the brain and autonomic nervous system.
- Neurotransmitter dopamine plays a role in the brain.
- Striving, focusing, and finding something interesting rely on it.
- It is associated with pleasure.

Parkinson's Disease:

1. Significantly low levels of DOPAMINE in the substantia nigra; death of dopaminergic neurons
2. Protein misfolding disease: alpha-synuclein misfolds (gains beta sheets) and polymerizes
3. Signs and symptoms: TRAP: tremors of the extremities, rigidity, akinesia, postural instability
4. Histological finding: Lewy bodies (alpha-synuclein fibrils) in the substantia nigra
5. Administration of L-DOPA (so it can be decarboxylated to dopamine, refer to the pathway)

Amino Acid-Derived Hormones



Tetrahydrobiopterin (BH4) is required by aromatic hydroxylases and NO synthases.

BH4 deficiency can cause hyperphenylalaninemia (PKU): Observed in *China, Turkey and Saudi Arabia*. *Inherited BH4 deficiency can be caused by a deficiency in dihydropteridine reductase which recycles BH4.*

Addison's Disease (please read notes!)

1. Adrenal insufficiency: $\downarrow\downarrow$ cortisol, $\downarrow\downarrow$ mineralocorticoids, $\uparrow\uparrow$ ACTH, $\downarrow\downarrow$ epinephrine; decreased expression of PNMT in chromaffin cells
2. Signs and symptoms include hypotension, fatigue, muscle weakness, weight loss, hyperpigmentation
3. **Autoimmune** – autoantibodies attack the adrenal gland

Failure of the adrenal cortex, as with **Addison's disease**, can suppress epinephrine secretion as the activity of the synthesizing enzyme, **phenylethanolamine-N-methyltransferase**, depends on the high concentration of **cortisol** that drains from the cortex to the medulla. **ACTH stimulates the adrenal cortex to release cortisol, which increases the expression of PNMT in chromaffin cells, enhancing adrenaline synthesis.**

This is most often done in response to stress. The sympathetic nervous system, acting *via* splanchnic nerves to the adrenal medulla, stimulates the release of adrenaline. Acetylcholine released by preganglionic sympathetic fibers of these nerves acts on nicotinic acetylcholine receptors, causing cell depolarization and an influx of calcium through voltage-gated calcium channels. Calcium triggers the exocytosis of chromaffin granules and, thus, the release of adrenaline (and noradrenaline) into the bloodstream.

Signs and symptoms of Addison's Disease:

1. Chronic fatigue, HYPOTENSION
2. Muscle weakness

3. Loss of appetite
4. Weight loss
5. Abdominal pain
6. Loss of appetite
7. Craving salty food
8. Irregular menstrual period
9. Loss of interest in sex

Catecholamine Degradation

Two enzymes:

Monoamine oxidase (MAO) requires FAD

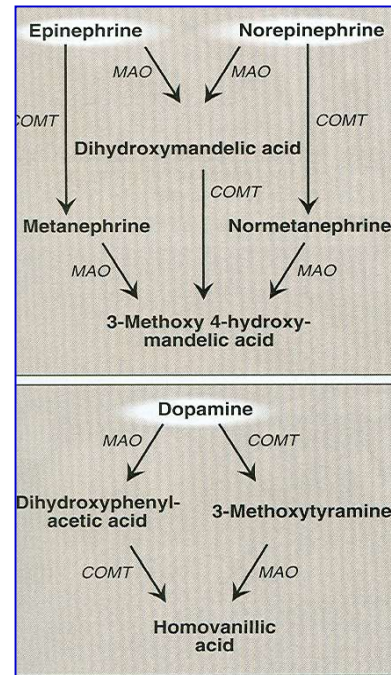
Catechol-O-methyl-transferase (COMT) requires SAM

Clinical Points:

1. MAO inhibitors are used as anti-depressants
2. Pheochromocytoma – tumor of catecholamine secreting cells

Symptoms: dangerous episodes of hypertension, diaphoresis, headaches, palpitations, arrhythmia, heart attack

Finding: ↑VMA (Vanillyl-mandellic acid)
↑metanephrine



Degradation of Catecholamines

- Monoamine oxidases inactivate catecholamines by deamination
- Catechol-O-methyl transferase methylates catecholamines

MAO has two subtypes found in neurons and astroglia:

MAO-A (also expressed in the liver, GI tract and placenta) deaminates serotonin, dopamine, epinephrine and norepinephrine;

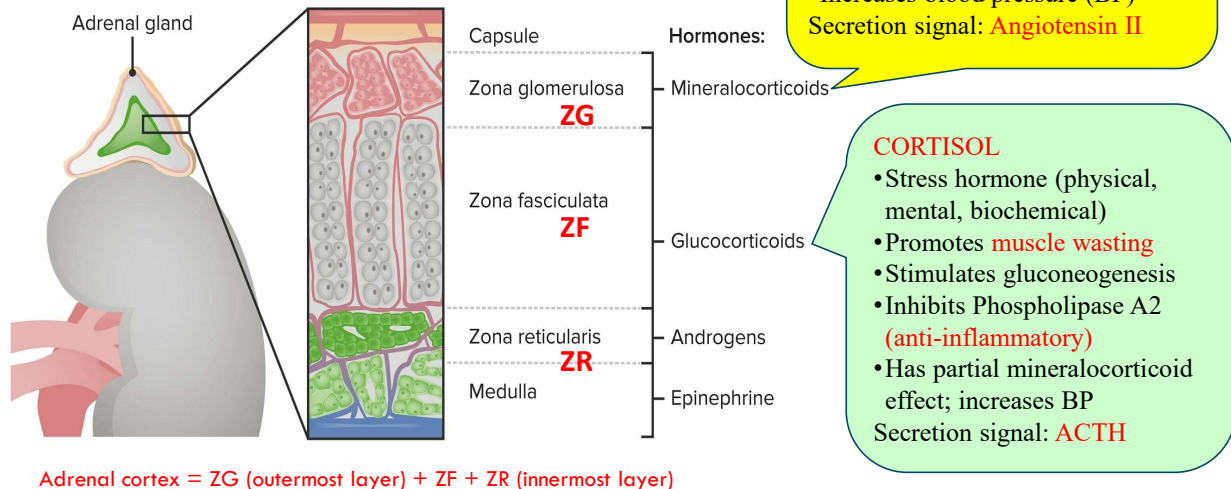
MAO-B (also expressed in blood platelets) degrade phenethylamine and dopamine.

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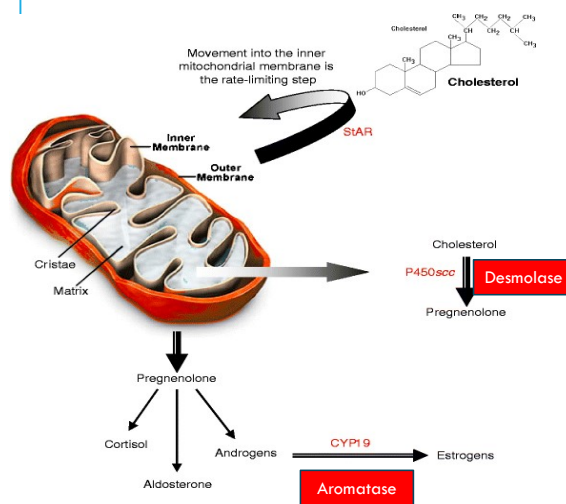
Steroid Hormones



Primary sections of the ADRENAL GLAND:

- 1. Adrenal medulla** (INNER portion): secretes epinephrine and to a lesser degree norepinephrine (hormones that work in conjunction with the sympathetic nervous system; flight or fight response)
- 2. Adrenal cortex** (OUTER portion-part of the hypothalamic-pituitary-adrenal axis): Secretes steroid hormones: mineralocorticoids (aldosterone), glucocorticoids (cortisol) and the androgens; all adrenal cortical hormones are synthesized from CHOLESTEROL

Steroidogenesis Rate-limiting Step



Rate-limiting step: **Steroidogenic Acute Regulatory Protein (StAR)** transports **cholesterol** from the outer to the inner mitochondrial membrane.

Present in the adrenal cortex, theca and luteal cells (ovary) and Leydig cells (testis).

StAR synthesis is stimulated by **LH, ACTH and angiotensin II**.

StAR synthesis is **cAMP-mediated**.

Enzymes in steroidogenesis: **CYP450**

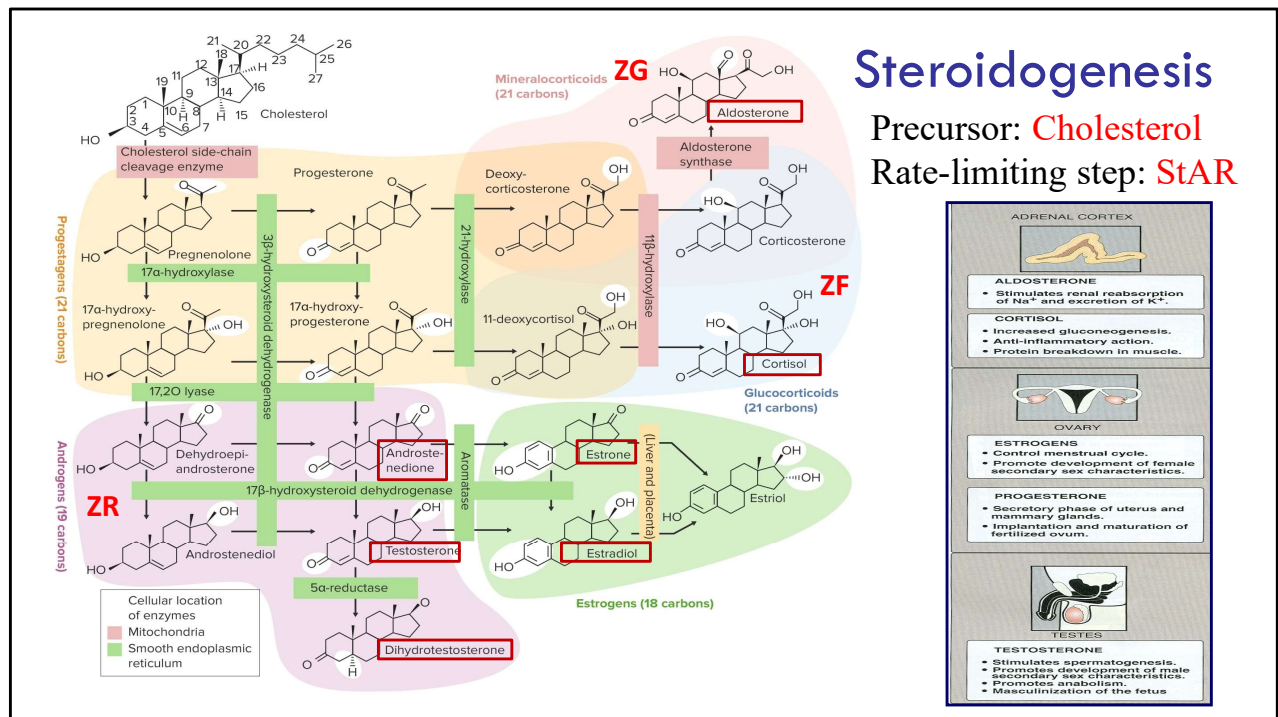
1. **heme-containing**
2. Requires **O₂** and **NADPH**

LH – leutenizing hormone

ACTH – adrenocorticotropic hormone

Regulation of StAR

- StAR is a mitochondrial protein that is rapidly synthesized in response to stimulation of the cell to produce steroid. Hormones that stimulate its production depend on the cell type and include luteinizing hormone (LH), ACTH and angiotensin II.
- At the cellular level, StAR is synthesized typically in response to activation of the cAMP second messenger system, although other systems can be involved even independently of cAMP.
- StAR has thus far been found in all tissues that can produce steroids, including the adrenal cortex, the gonads, the brain and the nonhuman placenta. One known exception is the human placenta.



Mineralocorticoids:

Synthesized in the adrenal cortex Zona Glomerulosa

Promotes sodium retention in the kidneys

Aldosterone (main mineralocorticoid)

Cortisol (however, cortisol doesn't typically function actively as a mineralocorticoid because it is rapidly oxidized in the kidney).

Cortisol is typically converted to cortisone by 11-beta-hydroxy steroid dehydrogenase rapidly in the kidney. This enzyme is inhibited by Glycyrrhizic acid, an active component of licorice. Overconsumption of licorice prevents conversion of cortisol, which then acts as a mineralocorticoid. This causes AME (Apparent Mineralocorticoid Excess). AME can result in hypertension, hypokalemia, alkalosis, and low plasma renin and low plasma aldosterone.

Glucocorticoids:

Synthesized in the Zona Fasciculata

Cortisol

Cortisone (from cortisol, nearly as potent as a glucocorticoid; acts by being converted back into Cortisol in the liver).

Sex Hormones:

Androgens

Synthesized in Zona Reticularis (some) and in gonads

Testosterone

Dihydrotestosterone (most potent)

Androstenedione

Estrogens

Synthesized mainly in the ovaries :

estradiol (most potent),

estrone, estriol

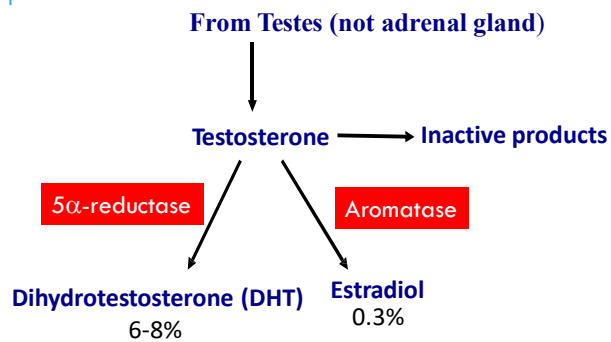
Congenital Adrenal Hyperplasia (CAH)

Deficiency in:	↑↑ Increase in:	↓↓ Decrease in:	Clinical Features
21-α-hydroxylase (most common) AR	Androgens K ⁺	Na ⁺ , Glucose Mineralocorticoids Glucocorticoids	Salt wasting, hypotension, hyperkalemia Male precocious puberty Masculinization of the female genitalia
11-β-hydroxylase AR	Androgens 11-deoxycortisol 11deoxycorticosterone	Glucose Aldosterone Cortisol	Low renin hypertension, fluid retention Male precocious puberty Masculinization of the female genitalia
17-α-hydroxylase AR	Mineralocorticoids Na ⁺	Sex hormones, K ⁺ Glucocorticoids	Low renin hypertension, fluid retention Hypogonadism All patients have female genitalia

Ambiguous genitalia (clitoral-like phallus), impotence, hair loss, gynecomastia are seen in **5- α -reductase deficiency**.

Ambiguous genitalia (clitoromegaly, labioscrotal fusion), amenorrhea, hirsutism, insulin resistance, tall stature, osteoporosis are seen in **aromatase deficiency**.

Fate of Testosterone in Males



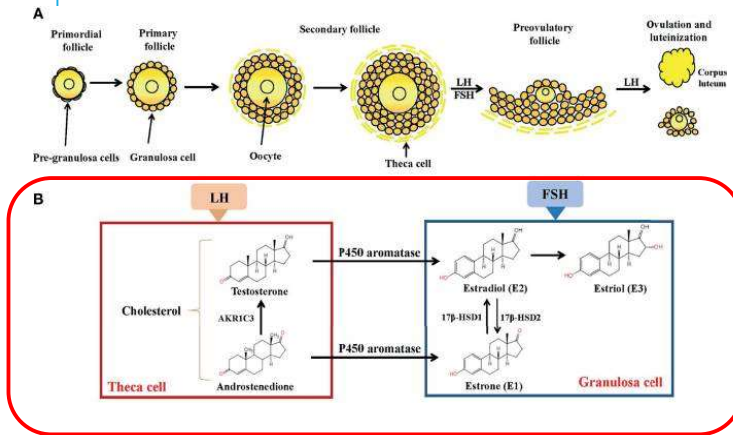
Androgen potency:
DHT > Testosterone > androstenedione

Link to Pharmacology:

Finasteride

- Inhibits 5 α -reductase
- $\downarrow\downarrow$ DHT
- Used in the treatment of benign prostatic hyperplasia (BPH) and alopecia
- What is a good molecular indicator to assess BPH?
 - $\uparrow$ PSA

Estrogen Synthesis in the Ovary



Link to Pharmacology:

Aromatase inhibitors are used in the treatment of estrogen-responsive breast cancer

In females, estrogen is produced by locally expressed P450 aromatase from follicular androgens.

Premenopausal women: E1 and E2 are primarily synthesized. Estradiol (E2) is the most potent and prevalent estrogen.

Menopausal women: Estrone (E1) is the most prevalent estrogen.

Estriol (E3) is produced by the placenta during pregnancy.

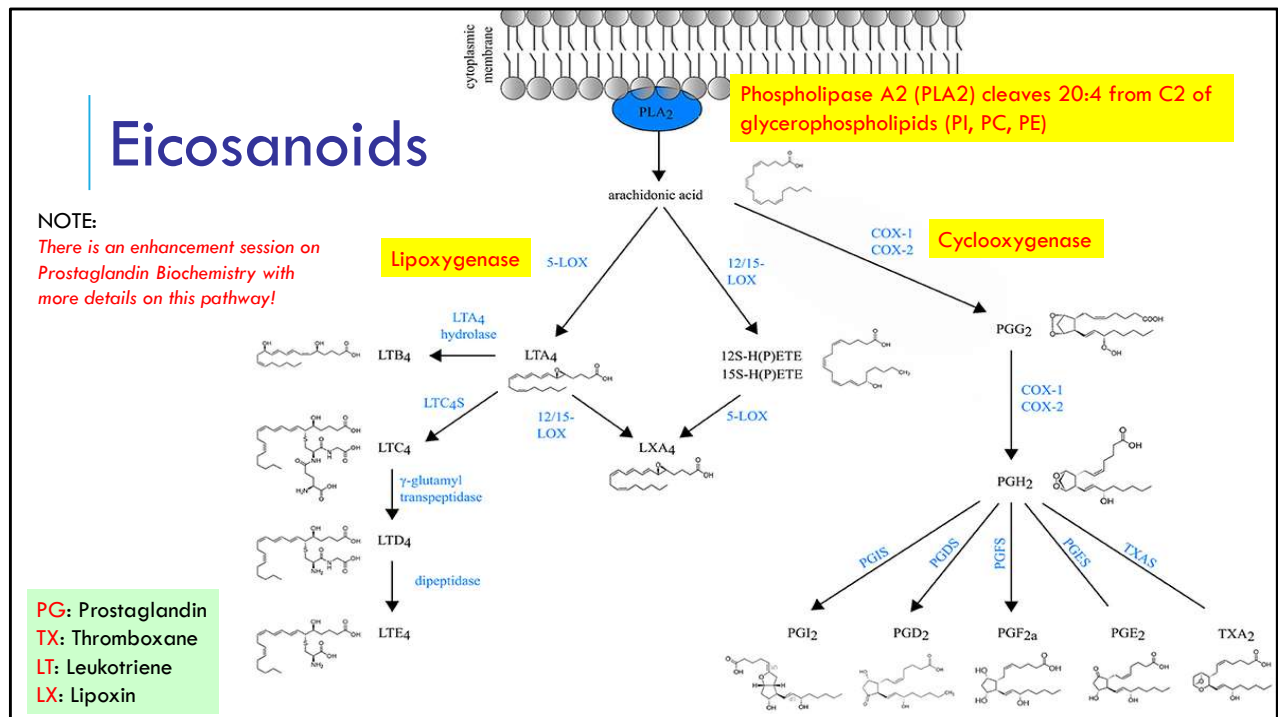
Estrogen potency:

Estradiol > estrone > estriol > estetrol (E4)

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- 1.Classes of hormones
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Schematic overview of eicosanoid biosynthesis:

Arachidonic acid released from membrane phospholipids by **cytosolic phospholipase A2** can be enzymatically converted either to prostaglandins (PGs) and thromboxane (TXA₂) by COX enzymes or to leukotrienes (LTs) and lipoxins (LXA₄s) by lipoxygenases (LOXs).

5-LOX: 5-lipoxygenase

12/15-LOX: 12/15-lipoxygenase

LTC4S: LTC4 synthase

PGIS: PGI or prostacyclin synthase

PGDS: PGD₂ synthase

PGFS: PGF synthase

PGES: PGE synthase

TXAS: TXA₂ synthase

Precursor of eicosanoids: **Arachidonic acid** (20:4, eicosatetraenoic acid)

Precursor of **arachidonic acid**: **Linoleic acid** (18:2, omega-6 FA)

- Eicosanoids are typically stored within cells.
- Synthesis is on-demand and is affected by physical, chemical and hormone stimuli.

Eicosanoids:

Prostaglandins: PG

Thromboxanes:TX (e.g. **TXA2** –strong platelet aggregation)

Leukotrienes:

HETE – hydroxyeicosatetraenoic acid

HPETE – hydroperoxyeicosatetraenoic acid

Critical enzymes:

1. PLA2- Phospholipase A2 releases arachidonic acid (20:4) from C2 of phospholipids like phosphatidyl inositol (PI) inhibited by SAIDs (steroidal anti-inflammatory drugs like cortisol and other glucocorticoids)
1. COX – cyclooxygenase – inhibited by NSAIDs (non-steroidal anti-inflammatory drugs, e.g. aspirin, ibuprofen)
2. LOX –lipoxygenase

Eicosanoids

Eicosanoids:

Prostaglandins: PG (via COX)

Thromboxanes: TX (via COX)

(e.g. **TXA₂** –strong platelet aggregation)

Leukotrienes: LT (via LIPOX)

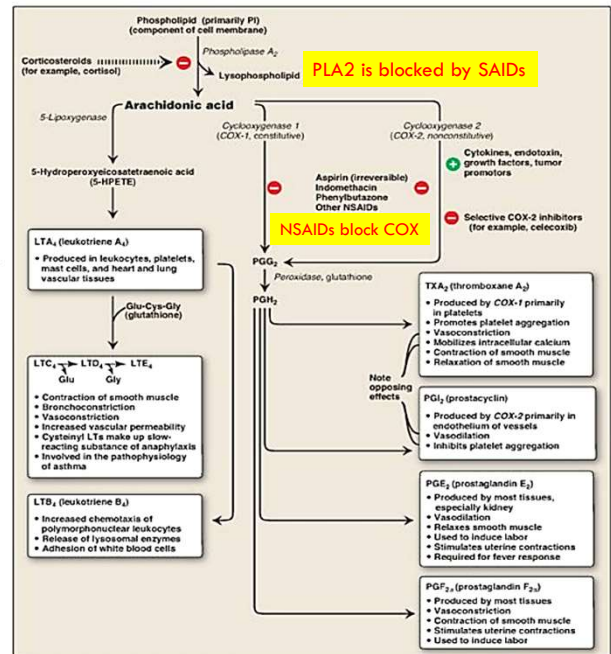
Lipoxins: anti-inflammatory; can be aspirin-induced (via LIPOX)

Clinical Points:

Steroidal Anti-Inflammatory Drugs (SAID) – glucocorticoids (e.g. cortisol) inhibit the release of arachidonic acid from C2 of PI, PE or PC by blocking Phospholipase A₂(PLA₂)

Non-Steroidal Anti-Inflammatory Drugs (NSAID)

- block cyclooxygenase
- Aspirin, ibuprofen, indomethacin, etc.



Good examples of Eicosanoids: (refer to the figure)

1. TXA₂ – has strong platelet aggregating property
2. PGI₂ – has anti-platelet aggregating property
3. LTC₄ to LTD₄ to LTE₄ – slow reacting substances of anaphylaxis (severe allergic response, bronchoconstriction, treatment: epinephrine)
4. Lipoxins – reduces excessive tissue injury and chronic inflammation

Eicosanoids are signaling molecules made by the enzymatic or non-enzymatic oxidation of arachidonic acid or other polyunsaturated fatty acids (PUFAs) that are, similar to arachidonic acid, 20 carbon units in length.

Eicosanoids are a sub-category of oxylipins, i.e., oxidized fatty acids of diverse carbon units in length, and are distinguished from other oxylipins by their overwhelming importance as cell signaling molecules.

Eicosanoids function in diverse physiological systems and pathological processes

such as: mounting or inhibiting inflammation, allergy, fever and other immune responses; regulating the abortion of pregnancy and normal childbirth; contributing to the perception of pain; regulating cell growth; controlling blood pressure; and modulating the regional flow of blood to tissues.

In performing these roles, eicosanoids most often act as autocrine signaling agents to impact their cells of origin or as paracrine signaling agents to impact cells in the proximity of their cells of origin.

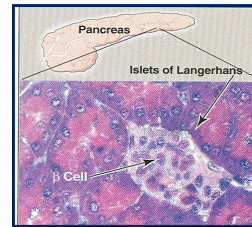
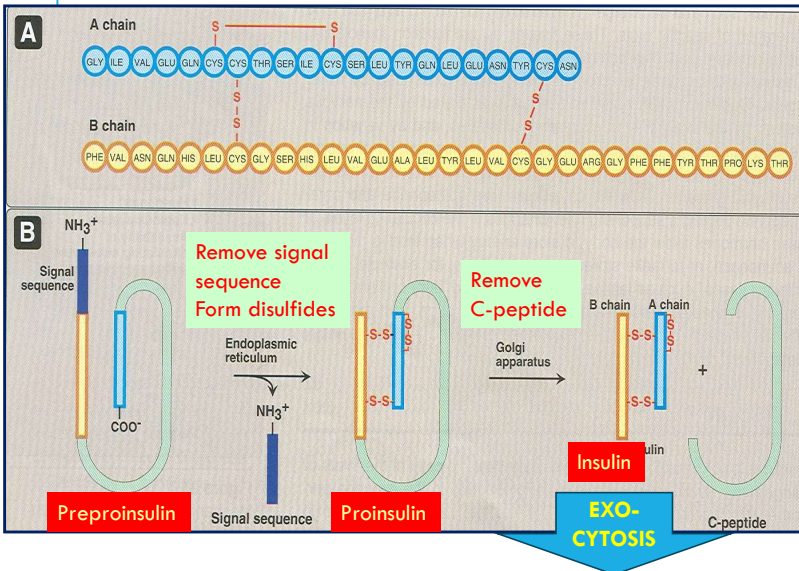
Eicosanoids may also act as endocrine agents to control the function of distant cells. PGA₂ (Prostaglandin A₂), is an activator of intrinsic apoptotic pathways by affecting mitochondrial membrane permeability.

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Insulin (Peptide Hormone)



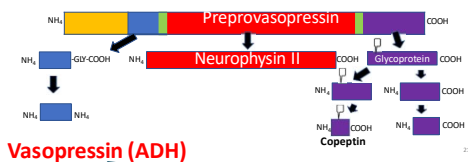
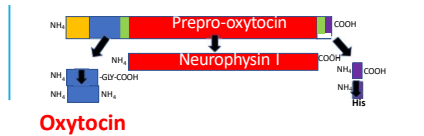
- Insulin has two chains:
 - alpha chain - 21 amino acids
 - beta chain - 30 amino acids
- 1 intra-strand disulfide within the A; 2 inter-strand disulfides between the A and B chains
- Insulin and C-peptide are packaged into secretory granules and secreted in equimolar amounts
- 5–15 % secretion is still proinsulin
- C-peptide (t_{1/2} = 30–35 mins) can be used as marker for diabetes and pancreatic cancer.

Insulin has a short plasma half-life (4–6 min), as would be expected from the necessity to respond rapidly to changes in blood glucose. C-peptide, on the other hand, has limited degradation in the liver and is degraded by the kidneys.

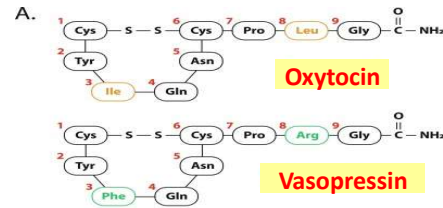
Hence, the half-life of C-peptide is around 30 to 35 minutes. As a result, although both insulin and C-peptide are secreted in equimolar concentrations, the molar ratio of circulating insulin to C-peptide is less than 1.

The Nobel Prize in Physiology or Medicine 1923 was awarded jointly to Frederick Grant Banting and John James Rickard Macleod "for the discovery of insulin."

Oxytocin and Vasopressin



Anti-diuretic hormone (ADH) stimulates the insertion of aquaporins (water channels) in the convoluted tubules increasing water permeability. In Syndrome of inappropriate anti-diuretic hormone (SIADH), water retention is abnormally increased with hyponatremia. In diabetes insipidus (AVP deficiency or AVP resistance), ADH is deficient (non-nephrogenic) or there is an ADH resistance (nephrogenic).



Oxytocin

1. stimulates the contractile cells around the ducts of the mammary glands to eject milk
2. stimulates contraction of smooth muscle in the pregnant uterus.

Vasopressin

1. decreases urine volume by **increasing reabsorption of water** by collecting ducts of the kidney
2. decreases rate of perspiration in dehydration
3. increases blood pressure by stimulating contraction of smooth muscle cells in the wall of arterioles.

Oxytocin and vasopressin are synthesized primarily in the hypothalamus. Stored and secreted by the posterior pituitary.

Oxytocin are both nonapeptides that differ only by 2 amino acids. **Receptors for these neuropeptides have a 7-helix motif.**

In the **hypothalamus, oxytocin is made** in magnocellular neurosecretory cells of the supraoptic and paraventricular nuclei and is **stored in Herring bodies at the axon terminals in the posterior pituitary.** It is then **released** into the blood from the posterior lobe (neurohypophysis) of the pituitary gland. Vasopressin (ADH) is produced by magnocellular neurosecretory neurons in the **paraventricular nucleus of hypothalamus (PVN) and supraoptic nucleus (SON).**

- **Oxytocin** is mature when the terminal glycine is removed leaving the amino group

- **Neurophysin I is the carrier protein for oxytocin** within the neuron
- **Vasopressin/ADH** is mature when the terminal glycine is removed leaving the amino group
- **Neurophysin II is the carrier protein for vasopressin/ADH** within the neuron

SIADH Symptoms:

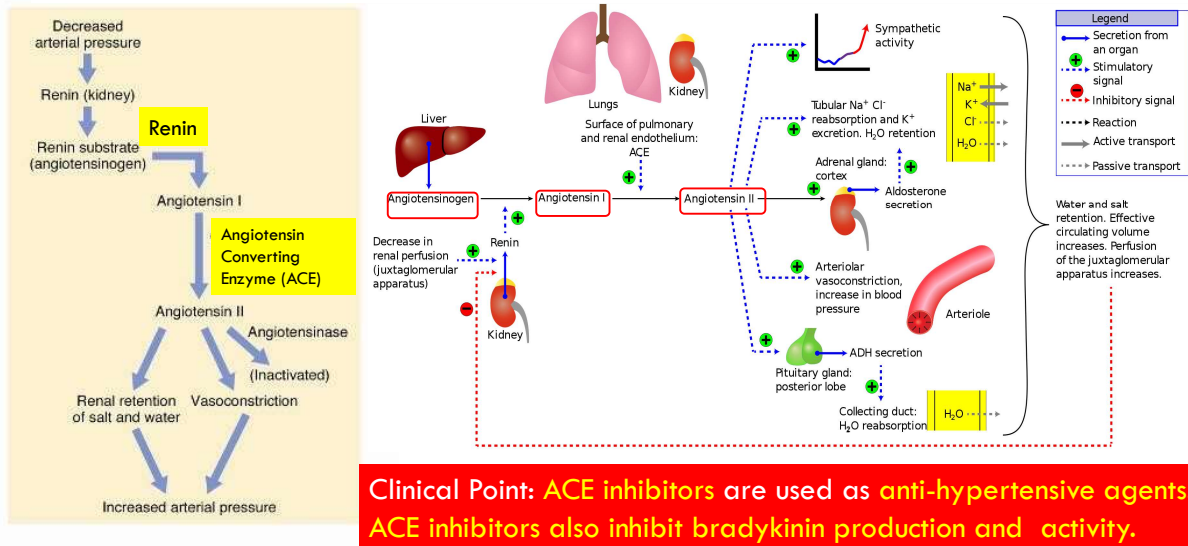
With SIADH, the urine is very concentrated. Not enough water is excreted and there is too much water in the blood. This dilutes many substances in the blood such as sodium. **Hyponatremia** (low plasma sodium) is the most common cause of symptoms of too much ADH. It is also the most common clue that a person may have SIADH.

Often, there are no symptoms from a mildly low sodium level. More symptoms occur the lower the level of the sodium.

When symptoms do occur, they may include any of the following:

- Nausea and vomiting
- Headache
- Problems with balance that may result in falls
- Mental changes, such as confusion, memory problems, strange behavior
- Seizures or coma if severe

Renin-Angiotensin System



Let's follow the system through and assume there is a drop in blood pressure. A drop in blood pressure → reduced arterial stretch → receptors 'fire' less. As result there is: Increased **sympathetic flow**; Decreased **parasympathetic flow**. Sympathetic activation → release of renin from juxta-glomerular cells. Renin then cleaves angiotensinogen (produced by the liver) to angiotensin I. Angiotensin I is converted in the lung to angiotensin II via the enzyme ACE. Angiotensin II, among its other functions, stimulates: The release of ADH and Aldosterone.

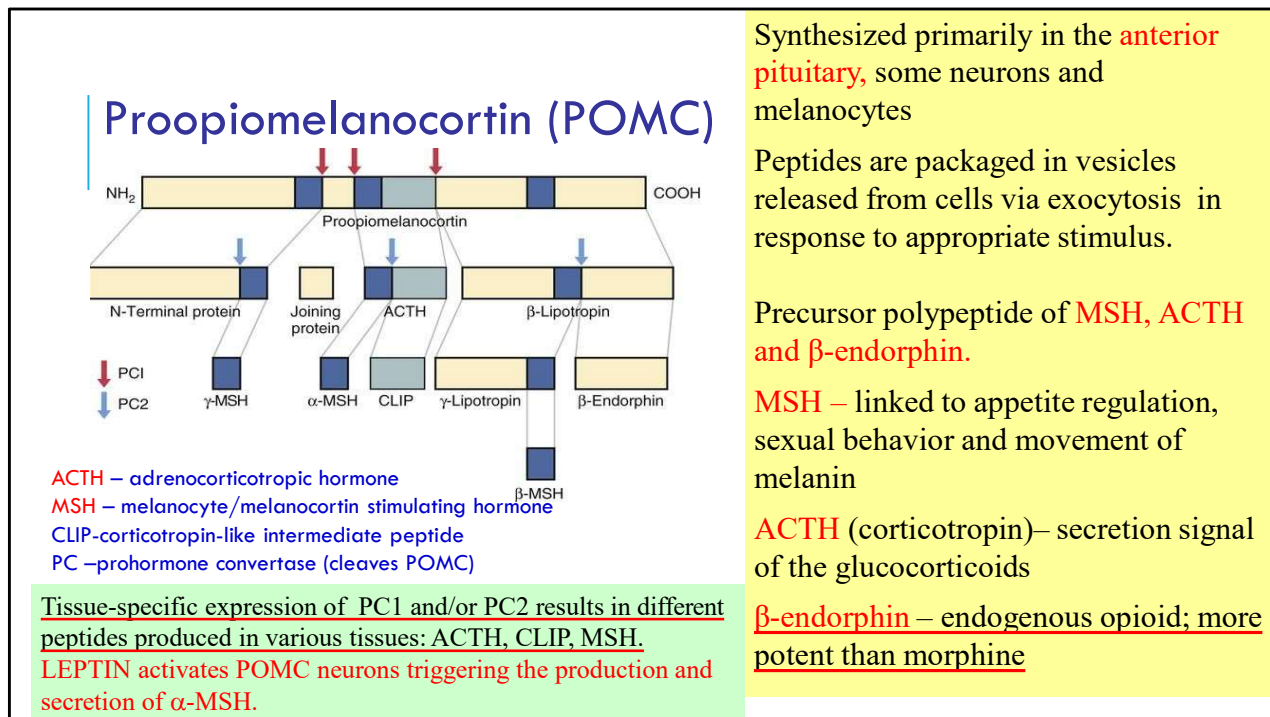
Aldosterone

1. Aldosterone is released from the zona glomerulosa, in response to: Renin, angiotensin II, high levels of potassium, ACTH and adrenergic stimulation.
2. Aldosterone causes sodium retention in exchange for excreting K⁺/H⁺ in the distal convoluted tubule.

Anti-diuretic hormone (ADH)

1. Synthesized by the hypothalamus (by the paraventricular and supraoptic neurons) and stored in the posterior pituitary.

2. Secreted In response to an increase in serum osmolarity (high levels of solutes and lower levels of water).
3. ADH acts on the distal convoluted tubule and collecting ducts within the kidneys binding to principal cells.
4. This triggers the vesicles to bind to the basolateral surface and apical membranes, to increase the number of aquaporin channels allowing more water (but not sodium) to be resorbed.
5. When serum osmolarity is low, less ADH is released and so less water is reabsorbed.



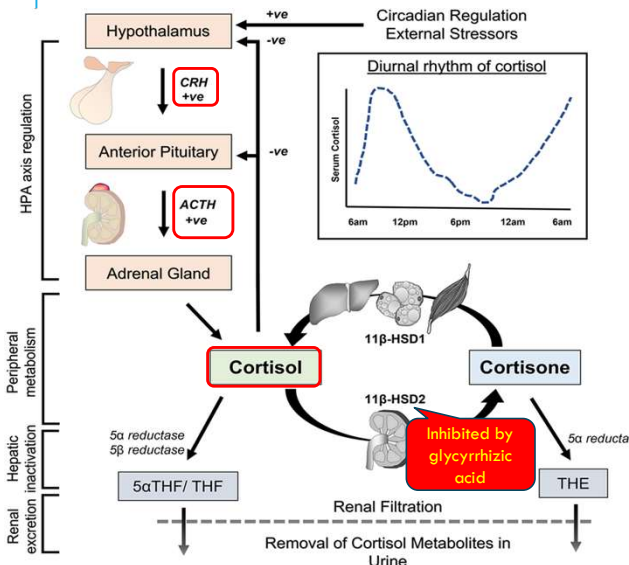
The POMC gene is actively transcribed in several tissues, including the corticotroph cells of the anterior pituitary, POMC neurons in the arcuate nucleus of the hypothalamus, cells of the dermis, and lymphoid tissue. In all of these cell types, POMC is processed to form a series of smaller peptides. The precise type of POMC-derived products from a particular tissue depends on the type of processing enzymes present in the tissue.

Thus, **pituitary corticotroph cells express *prohormone convertase 1 (PC1)* but not PC2**, resulting in the production of **N-terminal peptide, joining peptide, ACTH, and β-lipotropin**.

In the **hypothalamus**, the expression of **PC2** leads to the production of **α-MSH, β-MSH, γ-MSH, and β-endorphin**, but not ACTH. As discussed in lecture, α-MSH formed by neurons of the hypothalamus plays a major

role in appetite regulation.

Regulation of Cortisol Secretion



CRH: Corticotropin releasing hormone
ACTH: Adrenocorticotrophic hormone (corticotropin)

ACTH exerts its effect via cAMP.

Clinical Point: CUSHING SYNDROME

1. Hypercortisolism
2. Causes: Iatrogenic (exogenous), tumor of the pituitary ($\uparrow\uparrow$ ACTH, Cushing disease), tumor of the adrenal gland
3. Tests: 24-hour urinary free cortisol test
Dexamethasone suppression test
4. Buffalo hump, moon-faced, hirsutism and abnormal menses (females), central body obesity, ecchymosis, muscle atrophy, depression, hypernatremia, hypokalemia, hypervolemia, hypertension, hyperglycemia, immunosuppression

The hypothalamic-pituitary-adrenal (HPA) axis controls cortisol synthesis in the adrenal gland and its release into the circulation.

The activity of the HPA axis is set by the circadian clock and stress (physical trauma, infection, extremely cold and hot environment, physical exhaustion, mental anxiety)

Corticotropin-releasing hormone (CRH) is made and released from the hypothalamus and gives rise to ACTH release from the pituitary glands. CRH stimulates ACTH production in the anterior pituitary gland. Now ACTH sends signals to the adrenal gland (cortex), which secretes steroids (cortisol, androgen, and aldosterone). The rising cortisol level acts as a negative feedback mechanism and stops CRH and ACTH production.

Cortisol and inactive cortisone are interconverted by 11 β -hydroxysteroid dehydrogenase (11 β -HSD) enzymes for reversible activation and inactivation in peripheral tissues. Liver, fat and skeletal muscle are predominant sites of cortisol activation by 11 β -HSD1, while kidney is the predominant site for reversible cortisol inactivation by 11 β -HSD2. Terminal inactivation of cortisol occurs in the liver through enzymatic reduction to tetrahydrocortisol (THF), 5 α -tetrahydrocortisol (5 α -THF) and tetrahydrocortisone (THE). These metabolites are ultimately removed from the body through filtration in the kidneys and excretion in

urine.

CUSHING SYNDROME: Hypercortisolism

Causes of Cushing's Syndrome:

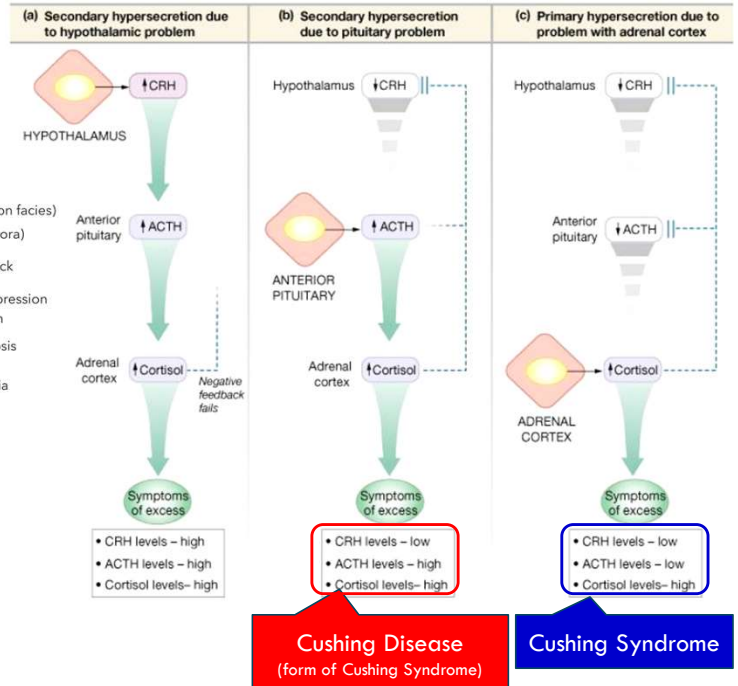
1. As a result of treatment with corticosteroids, this is the nonadrenal cause (Iatrogenic Cushing)
2. Secondary to a benign hormone-secreting pituitary adenoma. These are ACTH-secreting adenomas (68%).
 - a. Cortisol production from the adrenal tumors or nodules (17%).
 - b. Excess ACTH ectopic production (15%). Mostly there are malignant tumors.

Clinical presentation of Cushing's syndrome:

Patients have cardiovascular diseases. There is left ventricular hypertrophy.
High blood pressure changes were seen in 85 to 90%.
Central obesity is seen in 90%.
Glucose intolerance is seen in 80%.
Hirsutism is seen in 65%.
There are abnormal menses by 60%.
There may be muscle weakness in 60%.

Untreated cases have a 50% mortality for 5 years.

Hypercortisolism



Summary

- Hormones can be amino acid-derived, steroid-derived, lipid-derived, or be proteins or peptides
- Many, but not all, hormones circulate bound to carriers
- Tyrosine is the precursor of the catecholamines: dopamine, epinephrine, norepinephrine
- Tryptophan is the precursor of serotonin and melatonin
- Cholesterol is the basis for the steroid hormone family which include:
 - Mineralocorticoids, Glucocorticoids, Sex hormones (androgens and estrogens)
- Linoleic acid (18:2) is the precursor of arachidonic acid which gives the eicosanoids: lead to prostaglandins, thromboxanes, and leukotrienes
- Many hormones are synthesized by cleavage of longer peptides into shorter ones and may be stabilized by disulfide bonds

LET'S STUDY WITH ALL OUR MIGHT!

*Determination is the spark that
transforms challenges into victories.*



Drop a message for questions and clarifications:
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	EXPLORING OUR MOLECULAR MINDS <i>RECALL. REFLECT. VALIDATE.</i>
	Keyword Charts Practice Makes Perfect

KEYWORD CHART 1: *CAN YOU REMEMBER IMPORTANT INFO ABOUT THE FOLLOWING KEYWORDS?*

Keyword	Description/property/clinical correlate
Tetrahydrobiopterin (BH4)	
Steroidogenic Acute Regulatory Protein (StAR)	
Dihydrotestosterone (DHT)	
Angiotensin converting enzyme (ACE)	
Preproinsulin	
Adrenocorticotrophic hormone (ACTH)	

KEYWORD CHART 2

IDENTIFY THE DISORDER DESCRIBED IN THE SECOND COLUMN.

Disorder	Clinical/Biochemical Info
	Dangerous episodes of hypertension; diaphoresis; ↑↑Metanephrine ↑↑VMA
	clitoromegaly, labioscrotal fusion), amenorrhea, hirsutism, insulin resistance, hyperglycemia, tall stature, osteoporosis; decreased estrogens
	↑↑Phe ↓↓melanin ↓↓serotonin ↓↓dopamine
	Hypertension, hyperglycemia, buffalo-hump, hirsutism (women), abdominal striae, visceral fat deposition
	Masculinization of female genitalia, salt wasting, hypoglycemia, increased androgens
	Hypotension, fatigue, muscle weakness, weight loss, hyperpigmentation, ↑↑ACTH, ↓↓cortisol

Practice Makes Perfect!

A 7-year-old girl with masculinized genitalia was found to have a deficiency in 21-alpha-hydroxylase. Laboratory tests done on this girl's blood plasma will reveal which set of results?

	Aldosterone	Cortisol	Testosterone
I	↑	↑	↑
II	↓	↓	↓
III	↑	↑	↓
IV	↓	↓	↑
V	↑	↓	↓

Practice Makes Perfect!

A 35-year-old woman who suddenly develops an apple-type of body consults a physician due to a growing hump on her neck. She noticed that her arms and legs are getting very thin while her cheeks have become fuller and rounder. She complained of growing too much hair and acne on her face. A test that can confirm her condition is which of the following?

- A. Osmotic fragility test
- B. Van den Bergh reaction
- C. Prussian Blue staining
- D. Cytochrome oxidase staining
- E. Hemoglobin electrophoresis
- F. Dexamethasone suppression test

Practice Makes Perfect!

A 65-year-old man visits a urologist due to increased frequency of urination and an urgency to urinate especially at night. Upon examination, he was found to have an enlarged prostate gland. Finasteride was prescribed. Finasteride inhibits which of the following enzymes?

- A. 21-alpha-hydroxylase
- B. Aromatase
- C. 5- α -reductase
- D. 7-alpha-hydroxylase
- E. Lipoxygenase
- F. Cyclooxygenase

Practice Makes Perfect!

A 29-year-old pianist visits her physician because of difficulty sleeping over the past three weeks and has stayed away from her family and friends. In addition, she does not enjoy playing the piano as much as she used to. She has gained 4.5 kg in the past month. An inhibitor of which enzyme can be used to bring her mood up?

- A. Cyclo-oxygenase
- B. Monoamine oxidase
- C. Delta-aminolevulinate synthase
- D. Tyrosine hydroxylase
- E. Tryptophan hydroxylase

Practice Makes Perfect!

A 22-year-old woman sees her physician because of exhaustion and diarrhea. Physical examination shows that she has low blood pressure and blood studies show that she is anemic. Further examination revealed that she has a tumor of the pituitary gland. An intravenous drip of adreno-corticotrophic hormone (ACTH) fails to increase the concentration of 17-hydroxycorticoids. The intracellular effects of ACTH is mediated via an increase in which second messenger?

- A. IP_3
- B. CRH
- C. ACE
- D. cAMP
- E. cGMP